

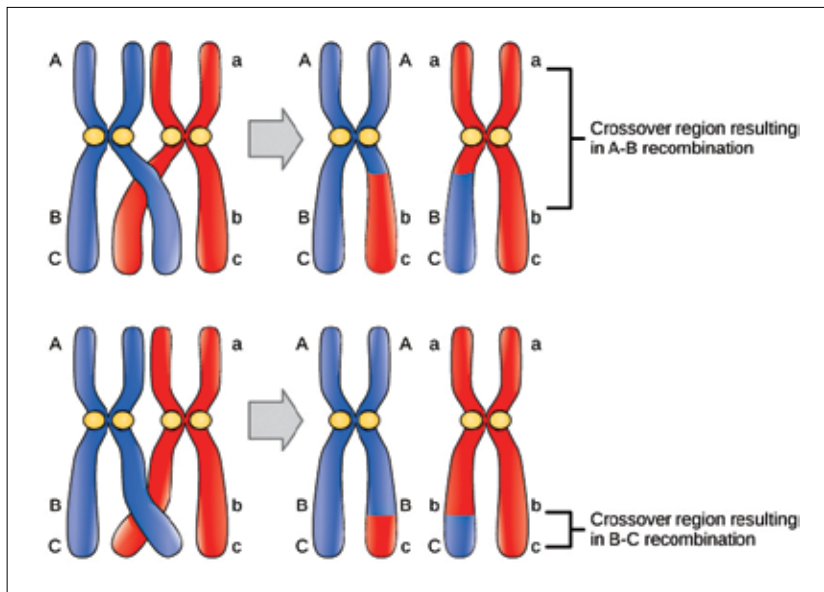
Inverse Chladni patterns

This mesmerising experiment shows the strange reaction of vibrating glass beads underwater. <https://digit.in/june19-50>



Dino crawled before it walked

Learning to crawl before learning to walk is actually a very unusual trait among animals. <https://digit.in/june19-51>



Crossover may occur in different locations of a chromosome.

gametes (reproductive cells). The closer genes were on a chromosome, the higher the chance of getting inherited together and genes would get separated during recombination if they were farther away from each other on a chromosome. This became the basis for the construction of genome maps.

FIRST GENETIC MAPS

Soon after Morgan presented his hypothesis, his student Alfred Sturtevant, began mapping out the genes on a chromosome. This became the first real-world application of gene mapping. However, he conducted gene mapping on fruit flies. Through his experiment, he gathered that double recombination can occur if the genes are placed far apart on the chromosome. He also deduced that initial recombination can inhibit further events of recombination by a phenomenon called interference. Gene mapping was then conducted in dogs and other model organisms like cows, pigs and insects.

HUMAN GENOME MAPS

Gene mapping in humans did not begin until the 1950s since it was exceptionally hard to understand and know what traits were actually caused by genetic mutations. In the 1960s, human genes were being mapped in autosomes

which do not have any sex traits. RFLPs (Restriction fragment length polymorphism) was described in 1980 as differences among individuals in the length of DNA fragments which were cut by enzymes. This initiated a large effort to generate maps of all chromosomes. The first human chromosome maps were created in the early 1980s, however, they only mapped some parts of the chromosomes and a few markers.

Maps of entire chromosomes were finally made by late 1980s which could help locate genes responsible for diseases. Whole-genome genetic maps were made available on the internet as well by mid-1990s.

Maps of entire chromosomes allowed for genome sequencing. A genome sequence actually pinpoints the order of every DNA base in a genome. The very first finished, full length sequence of a human chromosome was produced in 1999, Chromosome 22 being the chosen chromosome to be sequenced. This chromosome was chosen because of its relatively smaller size and additionally, a detailed gene mapped was already available for this chromosome.

HUMAN GENOME PROJECT

The Human Genome Project, which began in 1990, was a 15-year long, collaborative research project with the goal

to understand and map all the genes in the human body accurately. The main goals were fleshed out by the National Academy of Sciences in the US and were later adopted by the National Institutes of Health and the Department of Energy. The HGP (Human Genome Project) researchers understood and mapped the human genome in three ways – creating maps to show locations of genes in the major markers of our chromosomes, determining the sequences of all the bases in our genes' DNA, and producing linkage maps which allow tracking of inherited traits (like genetic disease).

The HGP revealed that there are around 20,500 human genes and 3.2 billion letters in the human genome. The International Human Genome Sequencing Consortium published the preliminary draft of the human genome in 2001 with 90 per cent of the human genome successfully mapped. Surprisingly, the actual number of human genes was significantly fewer than estimates in the past, which ranged from 50,000 to 1,40,000. The finished human genome sequence was published in 2003 which was at least 99.99 per cent accurate. After completion of the HGP, identification of genome sequences in other model organisms were taken up more extensively.

APPLICATIONS AND THE FUTURE

Advanced methods that could widely disseminate the information that was generated from the HGP to physicians, scientists and others, are necessary to ensure rapid imbibing of these research results for the benefit of humanity. Biomedical technology and research are the biggest beneficiaries of the HGP. Researchers have taken huge strides in the study of polygenic and other complex human diseases, due to the human genome sequence.

Gene mapping, which led to genome sequencing, has and will continue to dramatically change the practice of medicine by allowing of new and targeted drugs. It will allow improving and individualising treatment plans which can help fight genetic diseases in a much more efficient manner. 